Semiparametric Decomposition of Dengue Transmission Dynamics in Singapore: A Bayesian Renewal Equation Model with Gaussian Process Residuals

Ansel Lim

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Abstract

We develop a Bayesian renewal equation model for weekly dengue effective reproduction number (R_t) in Singapore (2012–2022), decomposing $\log(R_t)$ into additive components for climate covariates and a residual Gaussian process (GP). Using a Hilbert Space GP (HSGP) approximation with a Mat'ern 3/2 kernel, we find that climate (temperature and rainfall) explains less than 1% of R_t variance at weekly resolution, while the residual GP captures 99.5% of temporal variation. Post-hoc analysis of the GP residual against serotype surveillance data reveals that 5 of 7 serotype switch events are associated with elevated transmission (posterior probability > 0.75), with serotype diversity leading R_t elevation by approximately 10 months. Cross-kernel comparison demonstrates that conclusions are robust to kernel specification: all four kernels tested (Mat'ern 1/2, 3/2, 5/2, squared exponential) yield statistically indistinguishable fits (ΔELPD < 2). Out-of-sample forecasting reveals a fundamental limitation: the GP provides no predictive skill beyond a random walk baseline at any horizon. Branching process analysis shows that serotype switches create measurable windows of outbreak vulnerability, with 2022 exhibiting the most sustained transmission (18 weeks of $P(\text{sustain}) > 0.5$). Counterfactual analysis identifies the February 2020 DENV-3 emergence as the only switch producing clearly significant excess cases (+4,419; 95% CrI: 2,925 to 6,043), though this is confounded with COVID-19 behavioural changes. These results suggest that serotype-driven immunity dynamics are the dominant driver of dengue R_t variation in Singapore, but the GP framework has fundamental limitations for causal decomposition and operational forecasting.

1 Introduction

Dengue is the most widespread mosquito-borne viral disease globally, with an estimated 390 million infections annually [Bhatt et al., 2013]. Singapore, despite its advanced public health infrastructure, experiences recurrent dengue epidemics driven by the interplay of climate, vector ecology, population immunity, and the sequential dominance of four dengue serotypes (DENV-1–4) [Ang et al., 2018]. Understanding the relative contributions of these drivers to transmission intensity is essential for epidemic preparedness and intervention targeting.

The effective reproduction number, R_t—the average number of secondary infections generated by a single case at time t—is the central quantity for real-time epidemic assessment [Cori et al., 2013]. Standard approaches estimate R_t using either the Cori method with a single smoothing window, or flexible Bayesian frameworks such as EpiNow2 [Abbott et al., 2020] that use a Gaussian process (GP) to regularise temporal variation. However, these approaches conflate all sources of R_t variation into a single smooth function, precluding attribution to specific epidemiological drivers.

We address this by building a semiparametric model where $\log(R_t)$ is decomposed into an additive linear predictor for measured covariates (climate) and a residual GP that captures unmeasured temporal structure. This framework enables us to ask: conditional on observed climate variation, what residual structure remains in R_t, and does it align with serotype-driven immunity dynamics?

Specifically, this report addresses four questions:

1. How much of weekly R_t variation in Singapore is attributable to climate versus unmeasured drivers?
2. Does the residual GP temporal structure align with serotype replacement events?
3. Are the core findings robust to GP kernel specification?
4. Can the GP framework provide operationally useful forecasts?

We extend this core analysis with branching process analysis, variational autoencoder (VAE) comparison, spectral analysis, and counterfactual quantification.

2 Methods

2.1 Data Sources

Dengue cases. Weekly notified dengue case counts for Singapore (January 2012 – December 2022; $N = 574$ weeks) were obtained from the Ministry of Health Weekly Infectious Diseases Bulletin via data.gov.sg. Cases include both dengue fever and dengue haemorrhagic fever.

Climate covariates. Weekly mean temperature (°C) and total rainfall (mm) were obtained from the Meteostat API (Singapore Changi Airport, WMO 48698), standardised to unit variance, and lagged by 4 weeks to reflect the delay between climate conditions and case onset through the extrinsic incubation period and reporting pathway.

Serotype proportions. Monthly DENV-1–4 proportions (2013–2022) were derived from NEA sequencing surveillance. Raw proportions were smoothed using GAM multinomial logistic regression [Finch et al., 2025], from which dominant serotype identity, switch events, and Shannon

entropy ($H = -\sum_{i=1}^{4} p_i \log p_i$) were computed. Serotype data were used only in post-hoc analyses, not as model covariates, to avoid identifiability problems arising from the fact that serotype proportions are derived from the same case surveillance data used in the outcome.

2.2 Latent Process for R_t

The latent R_t process follows:

$$\log(R_t) = \mu + \beta_{\text{temp}} \cdot \text{temp}_{t-4} + \beta_{\text{rain}} \cdot \text{rain}_{t-4} + f_{\text{residual}}(t) \tag{1}$$

where μ is an intercept, β_{temp} and β_{rain} are regression coefficients for standardised climate covariates at 4-week lag, and $f_{\text{residual}}(t) \sim \mathcal{GP}(0, k_\nu(t, t'))$ is a zero-mean GP with Mat'ern $\nu = 3/2$ kernel:

$$k_{3/2}(t, t') = \alpha^2 \left(1 + \frac{\sqrt{3}\,|t - t'|}{\rho}\right) \exp\left(-\frac{\sqrt{3}\,|t - t'|}{\rho}\right) \tag{2}$$

where α is the marginal standard deviation (controlling amplitude of $\log(R_t)$ excursions) and ρ is the length scale in weeks (controlling the temporal correlation range). The Mat'ern 3/2 produces once-differentiable sample paths—smoother than the Mat'ern 1/2 (Ornstein–Uhlenbeck) but less smooth than the squared exponential (infinitely differentiable). This is the default kernel in EpiNow2 [Abbott et al., 2020] because it permits R_t to change relatively quickly while maintaining continuity.

2.3 HSGP Approximation

Exact GP inference requires $\mathcal{O}(N^3)$ computation for N time points. We use the Hilbert Space basis function approximation [Riutort-Mayol et al., 2023] with $M = 110$ basis functions:

$$f(t) \approx \sum_{j=1}^{M} \beta_j \cdot \phi_j(t), \quad \beta_j \sim \mathcal{N}\left(0, \sqrt{S(\lambda_j)}\right) \tag{3}$$

where $\phi_j(t) = (1/\sqrt{L}) \sin(j\pi(t + L)/(2L))$ are Laplacian eigenfunctions on the domain $[-L, L]$ with $L = 1.5 \times (\max(t) - \min(t))/2$, and $S(\omega)$ is the spectral density of the Mat'ern 3/2 kernel:

$$S(\omega) = \alpha^2 \cdot \frac{4(\sqrt{3}/\rho)^3}{((\sqrt{3}/\rho)^2 + \omega^2)^2} \tag{4}$$

This reduces computational complexity from $\mathcal{O}(N^3)$ to $\mathcal{O}(MN)$, making inference tractable in Stan [Carpenter et al., 2017] via HMC/NUTS. The choice of $M = 110$ ensures that the highest eigenfrequency exceeds the effective bandwidth of the GP's spectral density at the 5th percentile of the ρ prior.

2.4 Transmission Model

Expected cases follow the discretised renewal equation [Cori et al., 2013]:

$$\lambda_t = R_t \sum_{s=1}^{S} c_{t-s}\, w_s \tag{5}$$

where c_{t-s} are observed cases, w_s is the discretised generation interval PMF ($S = 6$ weeks), and the sum represents the total infectiousness at time t. Rather than fitting a single parametric distribution to published serial interval estimates, the generation interval was constructed by Monte Carlo convolution of its biological components [Chan and Johansson, 2012]: intrinsic incubation period ($T_{\text{IIP}} \sim \text{Gamma}(\text{mean} = 5.9\text{d}, \text{SD} = 1.5\text{d})$), viremic period ($T_{\text{vir}} \sim \text{Uniform}(0, 5\text{d})$), extrinsic incubation period ($T_{\text{EIP}} \sim \text{Gamma}(\text{mean} = 11\text{d at } 27°\text{C})$), and mosquito biting interval ($T_{\text{bite}} \sim \text{Exponential}(\text{mean} = 2\text{d})$). The first $S = 6$ weeks are treated as burn-in.

2.5 Observation Model

$$c_t \sim \text{NegBin}(\lambda_t, \phi) \tag{6}$$

with $\text{Var}(c_t) = \lambda_t + \lambda_t^2/\phi$, accommodating overdispersion from clustered transmission and spatially heterogeneous reporting. The model assumes constant case ascertainment, which is reasonable for the renewal equation framework since a constant reporting fraction cancels in the R_t calculation [Cori et al., 2013]. Implications of serotype-dependent ascertainment are discussed in Section 4.3.

2.6 Prior Specification

Weakly informative priors encode epidemiological plausibility without strong prescription:

- **Intercept:** $\mu \sim \mathcal{N}(0, 0.5)$, placing the prior median R_t at $\exp(0) = 1$ with 95% interval $[0.37, 2.72]$, spanning the plausible range for endemic dengue.
- **GP length scale:** $\log(\rho) \sim \mathcal{N}(\log(15), 0.5)$, median 15 weeks (95% interval: 5.6 to 40.0 weeks). The longer prior reflects that the residual GP needs only to capture variation not explained by covariates.
- **GP amplitude:** $\log(\alpha) \sim \mathcal{N}(-1.2, 0.5)$, median 0.30 on the $\log(R_t)$ scale, implying typical excursions of ± 0.3 to ± 0.8.
- **Overdispersion:** $\phi \sim \text{half-}\mathcal{N}(0, 10)$, permitting substantial overdispersion or near-Poisson behaviour.
- **Covariate coefficients:** $\beta \sim \mathcal{N}(0, 0.5)$, centered at no effect.

2.7 Inference and Model Comparison

Models were fitted using CmdStan 2.38.0 with 4 chains of 1,000 warmup + 1,000 sampling iterations. Convergence was assessed via $\hat{R} < 1.01$, bulk ESS > 400, tail ESS > 400, and zero divergent transitions. Model comparison used approximate leave-one-out cross-validation (LOO-CV) via Pareto-smoothed importance sampling [Vehtari et al., 2017].

2.8 Post-hoc Serotype Analysis

Rather than incorporating serotype data as an in-model covariate, we perform a post-hoc analysis of the residual GP against serotype dynamics. The residual GP $f_{\text{residual}}(t)$ captures R_t variation not explained by climate. If serotype replacement events increase R_t by creating large susceptible

pools, then f_{residual} should show a specific temporal profile around switch events: a rise near the switch, followed by gradual decline during stable dominance.

For each of 7 identified switch events (2013–2022), we extract the posterior distribution of monthly f_{residual} in a 6-month window before and 12 months after the switch. We also compute cross-correlation functions (CCF) between f_{residual} and Shannon entropy. This analysis is descriptive: with only 7 events, the sample size precludes formal hypothesis testing.

2.9 Extensions

Cross-kernel comparison. The core model (Equation 1) was refitted with Mat'ern 1/2, 3/2, 5/2, and squared exponential kernels to assess sensitivity of conclusions to smoothness assumptions. The only change is the spectral density function $S(\omega)$ in Equation 3; all other model components remain identical.

Out-of-sample forecasting. Rolling-window forecasts at 4, 8, and 13-week horizons were evaluated against two baselines—random walk (last observed R_t held constant) and $R_t = 1$ (naive endemic)—using the continuous ranked probability score (CRPS) and calibration metrics. Serotype Shannon entropy at 43-week lag was tested as an additional leading indicator.

Branching process analysis. The posterior R_t and overdispersion ϕ were used to compute time-varying extinction probability q and outbreak sustaining probability $P(\text{sustain}) = 1 - q$ under a negative binomial offspring distribution, connecting the R_t estimates to outbreak risk in the branching process framework.

VAE comparison. A sliding-window variational autoencoder (window = 26 weeks, latent dimension = 4, hidden = 32) was trained on log-transformed case counts with β-annealing over 200 epochs to test whether a purely data-driven approach discovers the GP residual structure without mechanistic modelling assumptions.

Spectral analysis. The posterior mean f_{residual} was analysed via Lomb–Scargle periodogram to identify dominant periodicities and compare them against theoretical predictions from multi-serotype dynamics.

Counterfactual analysis. Excess cases attributable to non-climate drivers were computed by comparing observed cases against a counterfactual $R_t^{cf} = \exp(\mu + f_{\text{climate}})$ that zeros out the residual GP, quantifying the case burden attributable to unmeasured drivers at each serotype switch.

3 Results

3.1 Core Model: Climate Explains Less Than 1% of R_t Variance

The fitted model shows excellent calibration, with posterior predictive 95% coverage of 97.7% and 80% coverage of 92.3% (Figure 2). All MCMC diagnostics are satisfactory: zero divergent transitions, all $\hat{R} < 1.01$, and minimum ESS > 800.

Climate effects on R_t are negligible at weekly resolution. A 1-SD increase in temperature multiplies R_t by 0.989 (95% CrI: 0.969, 1.009); a 1-SD increase in rainfall multiplies R_t by 1.003 (95% CrI: 0.986, 1.020). Neither effect is distinguishable from 1.0. Variance decomposition

confirms that climate covariates explain only 0.4% (median) of R_t variation, with the residual GP absorbing 99.6% (Figure 1).

This result requires careful interpretation. The GP's dominance is partly a structural consequence of model specification: with a short estimated length scale ($\hat{\rho} = 2.86$ weeks; 95% CrI: 2.09, 3.71), the GP can track week-to-week outbreak dynamics that climate covariates—operating on seasonal timescales—cannot explain. The finding that climate is negligible *at weekly resolution* does not preclude climate being an important driver at seasonal or interannual timescales; rather, weekly R_t variation is dominated by faster-acting processes.

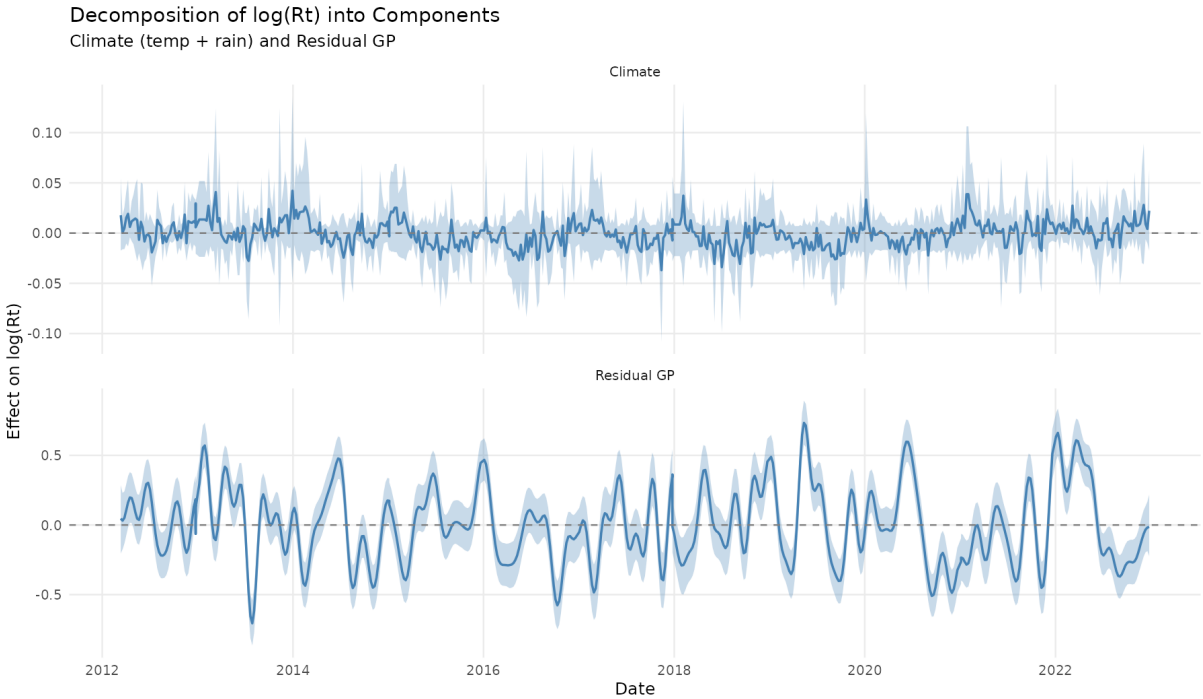

Figure 1: Decomposition of $\log(R_t)$ into climate and residual GP components. The residual GP (orange) captures nearly all temporal structure in R_t, while climate covariates (blue) contribute negligibly. Shaded bands show 95% credible intervals. The GP length scale of approximately 3 weeks allows tracking of individual outbreak dynamics.

3.2 Serotype Associations with the GP Residual

Post-hoc analysis of the GP residual against serotype surveillance data reveals consistent associations between serotype switch events and elevated transmission (Figure 3). Of 7 identified switch events (2013–2022), 5 show posterior probability > 0.75 of elevated f_{residual} in the 0–3 months post-switch.

The DENV-2→DENV-3 transition in February 2020 produces the strongest signal ($P = 1.0$; 20–35% R_t increase beyond climate prediction), consistent with a large naive susceptible pool encountering DENV-3 for the first time in many years. However, this event is temporally confounded with COVID-19 behavioural changes (more time spent at home may increase domestic *Aedes* exposure).

Cross-correlation analysis reveals that Shannon entropy leads f_{residual} by approximately 10–12 months (Figure 5), suggesting that rising serotype diversity precedes elevated transmission with

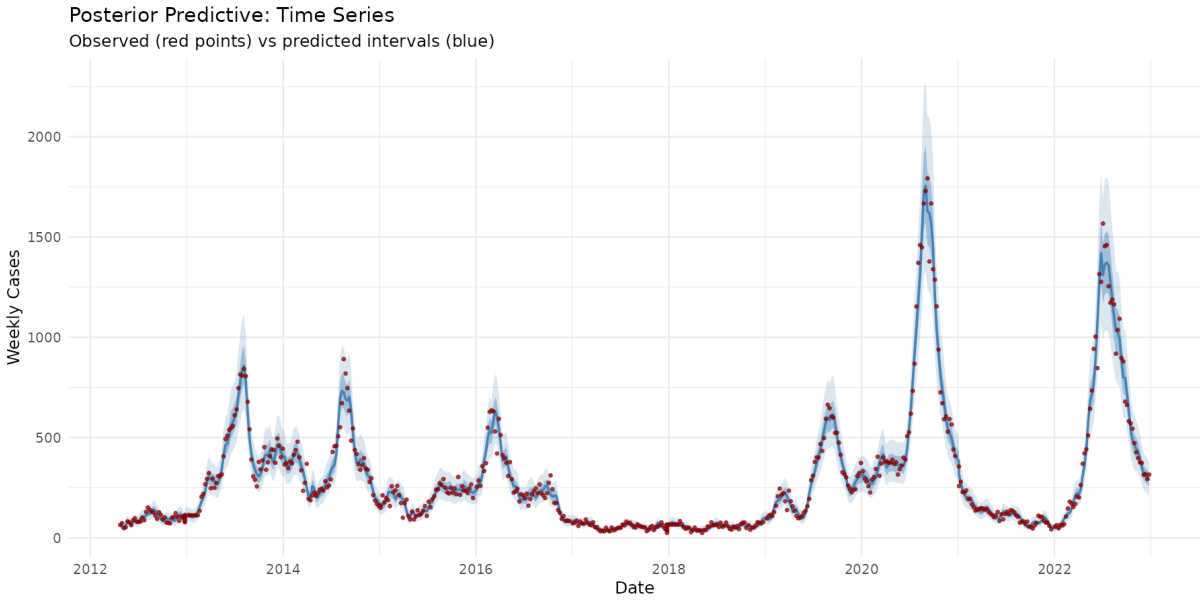

Figure 2: Posterior predictive check: observed weekly dengue cases (black points) against posterior predictive 95% intervals (shaded blue). Empirical coverage is 97.7% (nominal: 95%), indicating slight conservatism. The model captures both the timing and magnitude of major epidemic peaks (2013, 2019, 2020, 2022).

a lag consistent with the timescale of immunity waning and susceptible pool replenishment. This ~10-month lead is potentially actionable for seasonal risk assessment, though not for tactical weekly forecasting (see Section 3.4).

These associations are descriptive with $n = 7$ switch events, providing no meaningful sample size for formal hypothesis testing. The evidence is pattern-based: the temporal profiles match predictions from immunity-depletion theory in both direction (positive post-switch) and timing (aligned with observed serotype transitions), but cannot exclude alternative explanations.

3.3 Cross-Kernel Comparison: Conclusions Are Robust

All four kernels produce statistically indistinguishable fits, with $\Delta\text{ELPD} < 2$ and overlapping standard errors (Figure 6). The roughest kernel (Mat'ern 1/2) is marginally preferred (ELPD $= -2839.4$), consistent with dengue R_t exhibiting abrupt changes. The Mat'ern 3/2 is second ($\Delta\text{ELPD} = -0.8$, SE $= 0.5$), followed by the squared exponential ($\Delta\text{ELPD} = -1.6$, SE $= 0.7$) and Mat'ern 5/2 ($\Delta\text{ELPD} = -1.8$, SE $= 0.6$).

MCMC diagnostics are clean across all kernels: zero divergences, $\hat{R} \leq 1.01$, minimum bulk ESS > 805. Crucially, the serotype detectability pattern is preserved across kernels, with all four detecting the same 5/7 elevated-probability switches (Figure 7). This robustness confirms that the choice of Mat'ern 3/2 is defensible but not critical—all scientific conclusions hold regardless of kernel specification.

3.4 Forecasting: The GP Provides No Predictive Skill

The GP never beats a random walk baseline at any forecast horizon (Figure 8). At 4 weeks, CRPS skill versus random walk is -7.6% (the GP is *worse*); at 8 weeks, -2.3%; at 13 weeks,

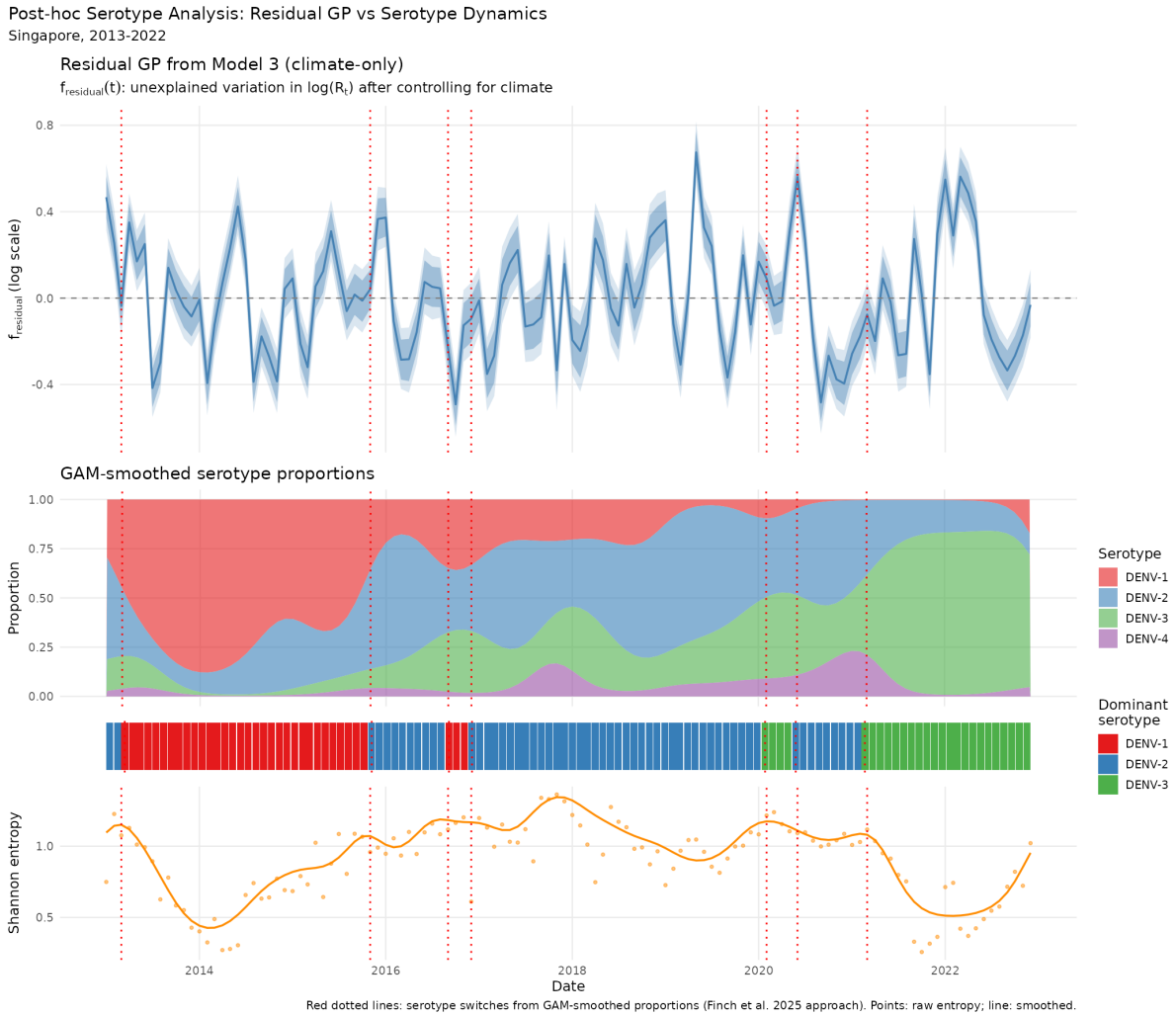

Figure 3: Three-panel serotype analysis showing temporal alignment between the GP residual and serotype dynamics. **Top:** Posterior median f_{residual} with 80% (dark) and 95% (light) credible bands. **Middle:** GAM-smoothed serotype proportions (DENV-1 red, DENV-2 blue, DENV-3 green, DENV-4 purple). **Bottom:** Shannon entropy computed from smoothed proportions. Vertical dashed lines mark serotype switch events. Five of 7 switches coincide with positive f_{residual} excursions.

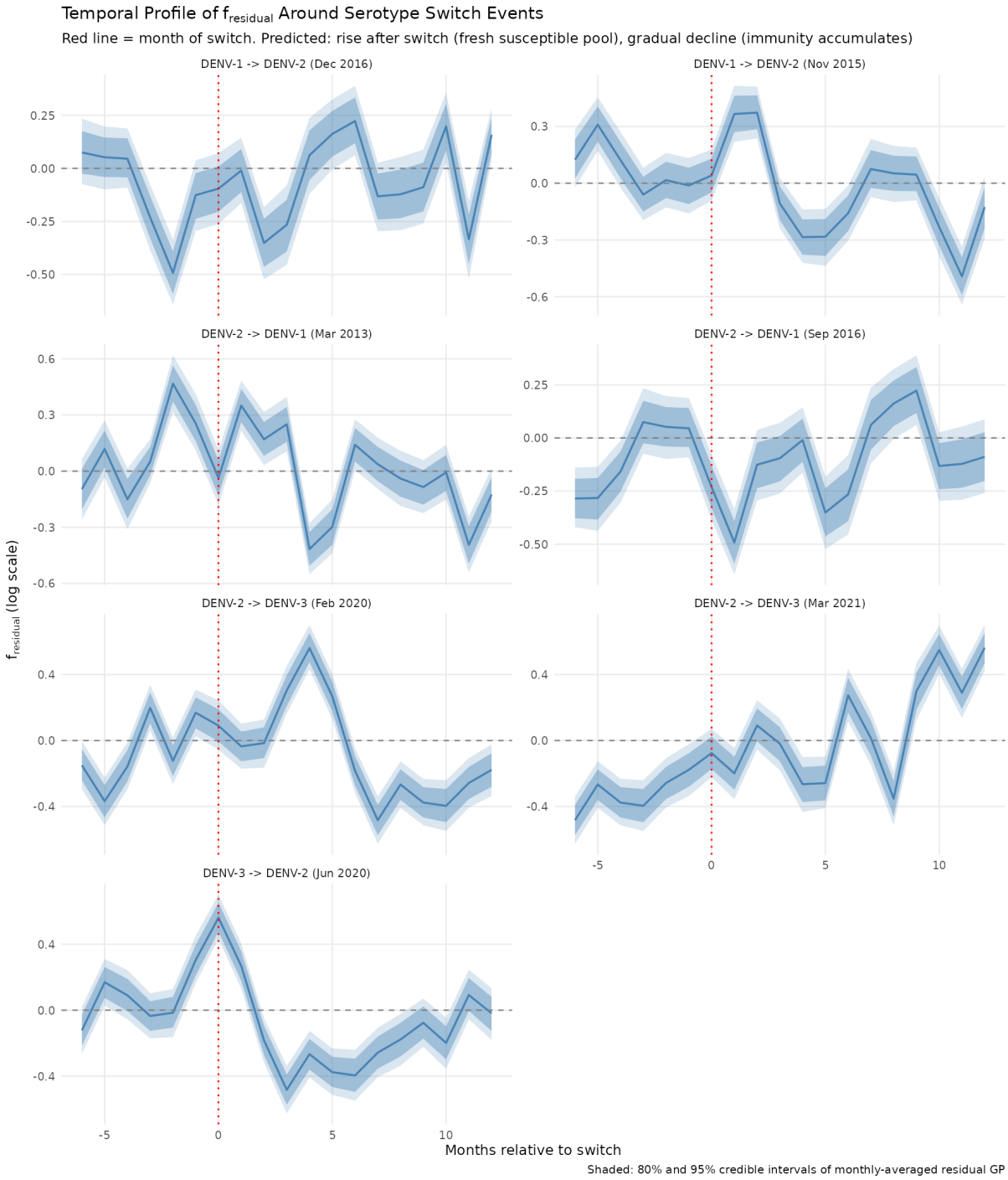

Figure 4: Temporal profiles of f_{residual} centred on each serotype switch event (time zero). Each panel shows one switch, with posterior median (line) and 95% credible band (shading). The predicted immunity-depletion pattern—rise at switch, gradual decline during stable dominance—is visible in most events, most clearly for the 2020 DENV-3 emergence.

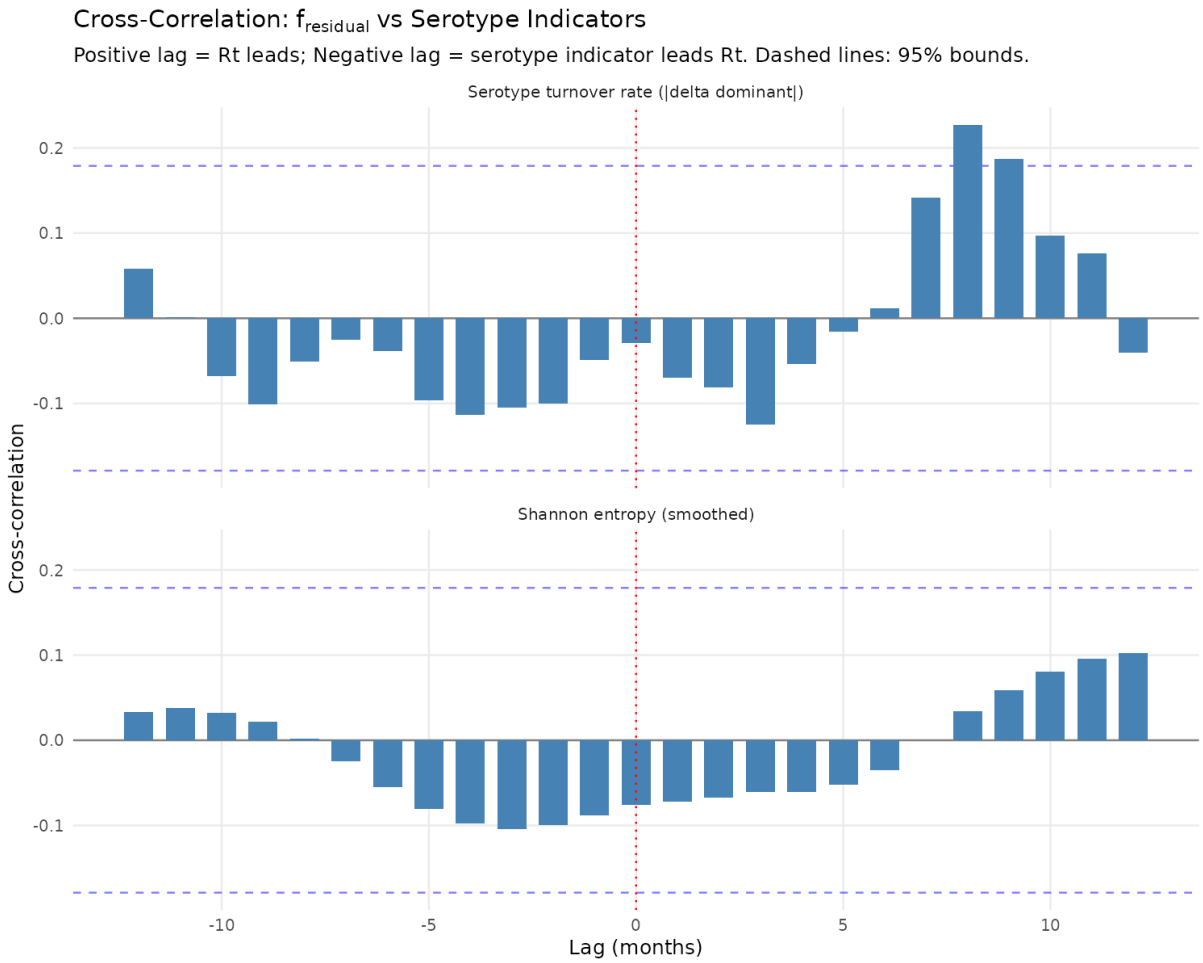

Figure 5: Cross-correlation function between monthly f_{residual} and Shannon entropy. Positive lags indicate entropy leading. Peak correlation at lag $+$10–12 months suggests that rising serotype diversity precedes elevated transmission with a lag consistent with susceptible pool dynamics.

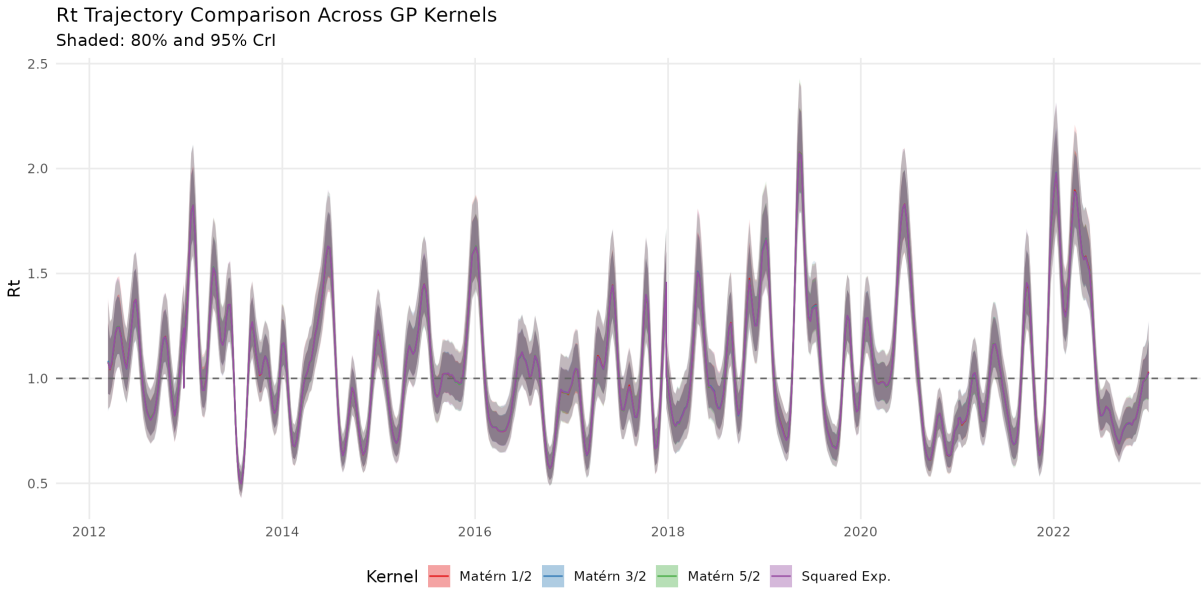

Figure 6: Overlaid R_t trajectories from all four kernel specifications: Mat'ern 1/2 (blue), Mat'ern 3/2 (orange), Mat'ern 5/2 (green), and squared exponential (red). Posterior medians are nearly identical; differences are smaller than credible interval widths. ΔELPD values are shown in the legend.

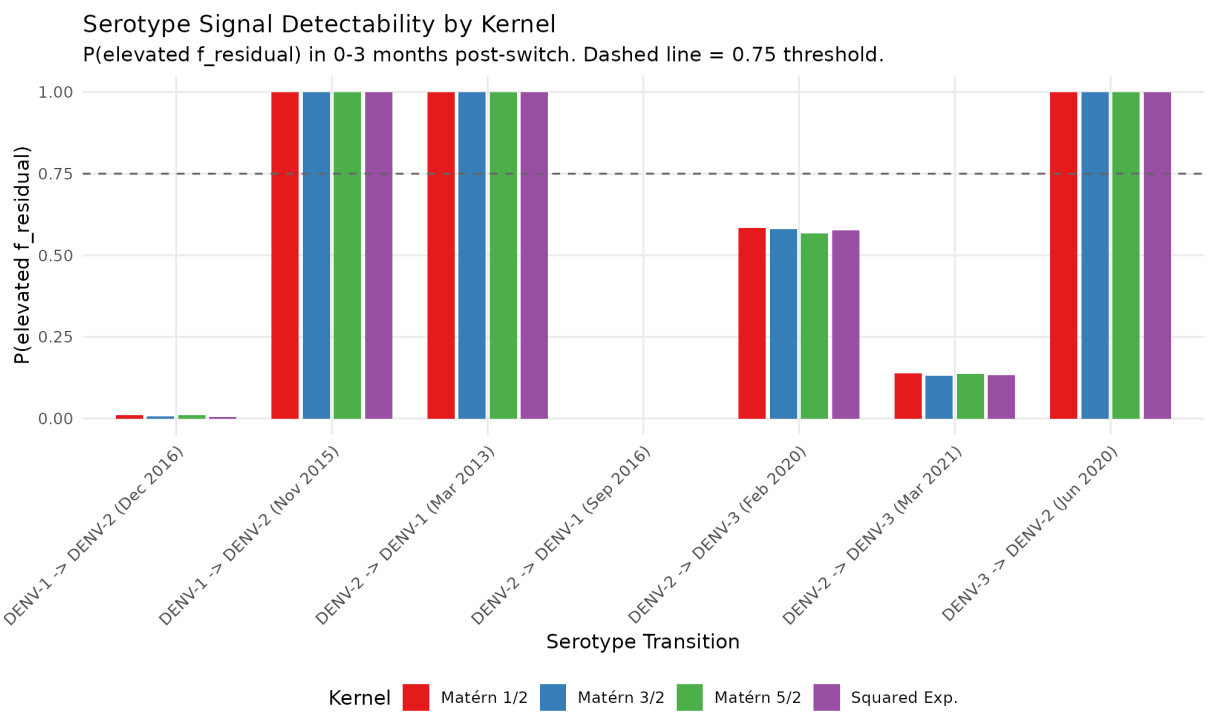

Figure 7: Serotype switch detectability across kernel specifications. Each bar shows $P(\text{elevated } f_{\text{residual}})$ in the 0–3 months post-switch. All four kernels identify the same 5 of 7 switches (horizontal dashed line at $P = 0.75$), confirming that serotype–R_t associations are not an artefact of the Mat'ern 3/2 assumption.

-10.5%. Calibration is conservative: 95% forecast intervals achieve 100% empirical coverage, indicating overly wide prediction bands rather than good calibration.

The apparent 35% skill versus the $R_t = 1$ baseline is misleading—both the GP and the baseline converge to similar predictions because the GP posterior regresses to the prior mean at horizons exceeding $\rho \approx 3$ weeks (Figure 9). Serotype entropy as a leading indicator does not improve forecasts at any horizon; the ~10-month leading relationship operates on a timescale too long for tactical weekly forecasting.

This is a fundamental, not incidental, limitation. The GP is a smoothing prior that regularises temporal variation retrospectively; it contains no mechanistic forward model and therefore provides no information about future trajectory beyond mean reversion. For operational dengue forecasting, models with explicit vector dynamics and immunity tracking are likely necessary.

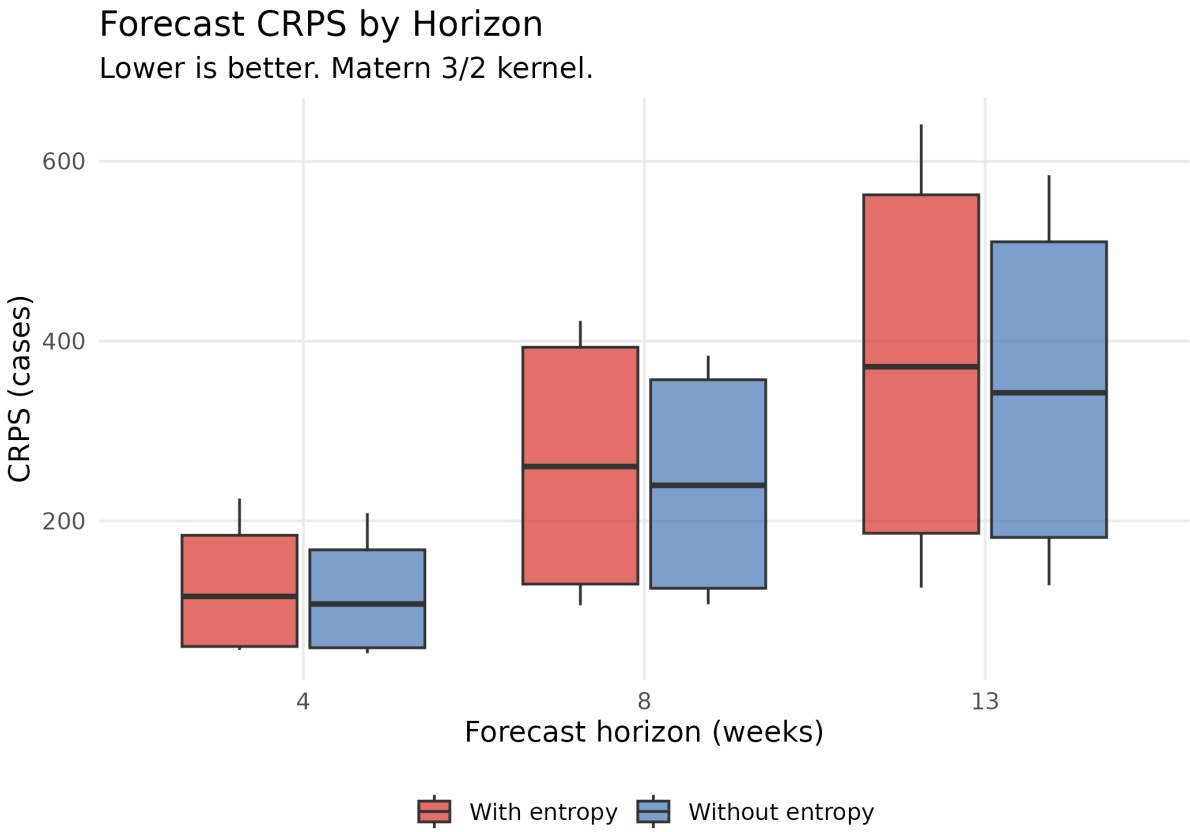

Figure 8: CRPS comparison across forecast horizons for the GP model (blue) versus random walk baseline (orange) and $R_t = 1$ baseline (green). Lower CRPS indicates better probabilistic forecast accuracy. The GP performs worse than the random walk at all horizons, confirming that its retrospective smoothing does not translate to predictive skill.

3.5 Branching Process: Serotype Switches Create Vulnerability Windows

The posterior R_t and overdispersion parameter yield a strong anticorrelation between R_t and extinction probability ($\rho_{\text{Spearman}} = -0.94$; Figure 10). Five of 7 serotype switches show increased outbreak sustaining probability ($P(\text{sustain})$) after the switch, with a mean increase of 6 percentage points.

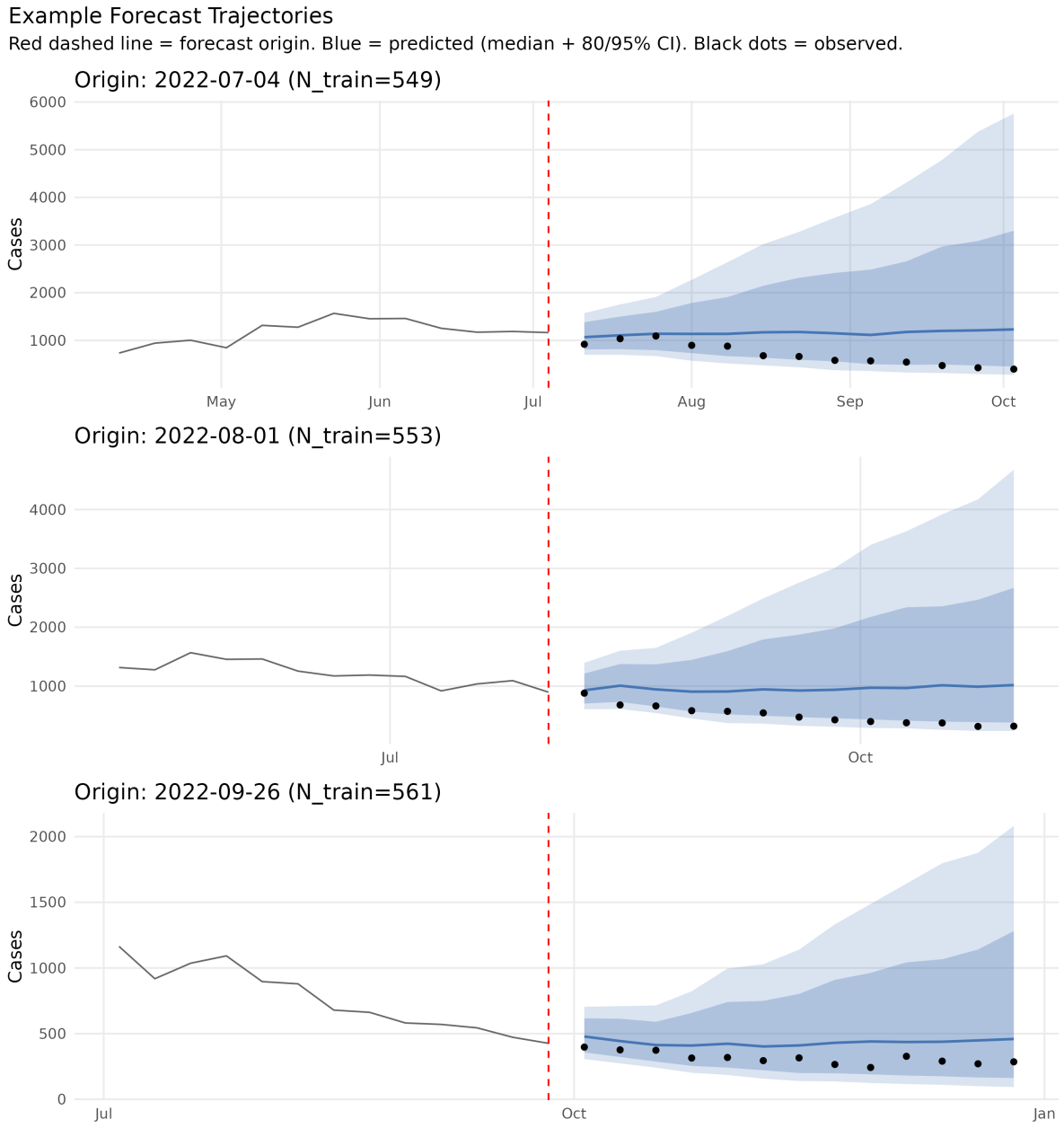

Figure 9: Example forecast trajectories at selected forecast origins, showing the GP posterior median (solid) and 95% prediction interval (shaded) diverging from observed R_t (dashed). The GP rapidly regresses to the prior mean ($R_t \approx 1$) within 3–4 weeks, consistent with the estimated length scale $\hat{\rho} \approx 3$ weeks.

The year 2022 stands out with the most sustained transmission: 18 weeks of $P(\text{sustain}) > 0.5$ and mean $R_t = 1.17$. This coincides with a period of DENV-3 circulation and relaxation of COVID-19 restrictions, illustrating the confluence of serotype-driven immunity depletion and behavioural changes.

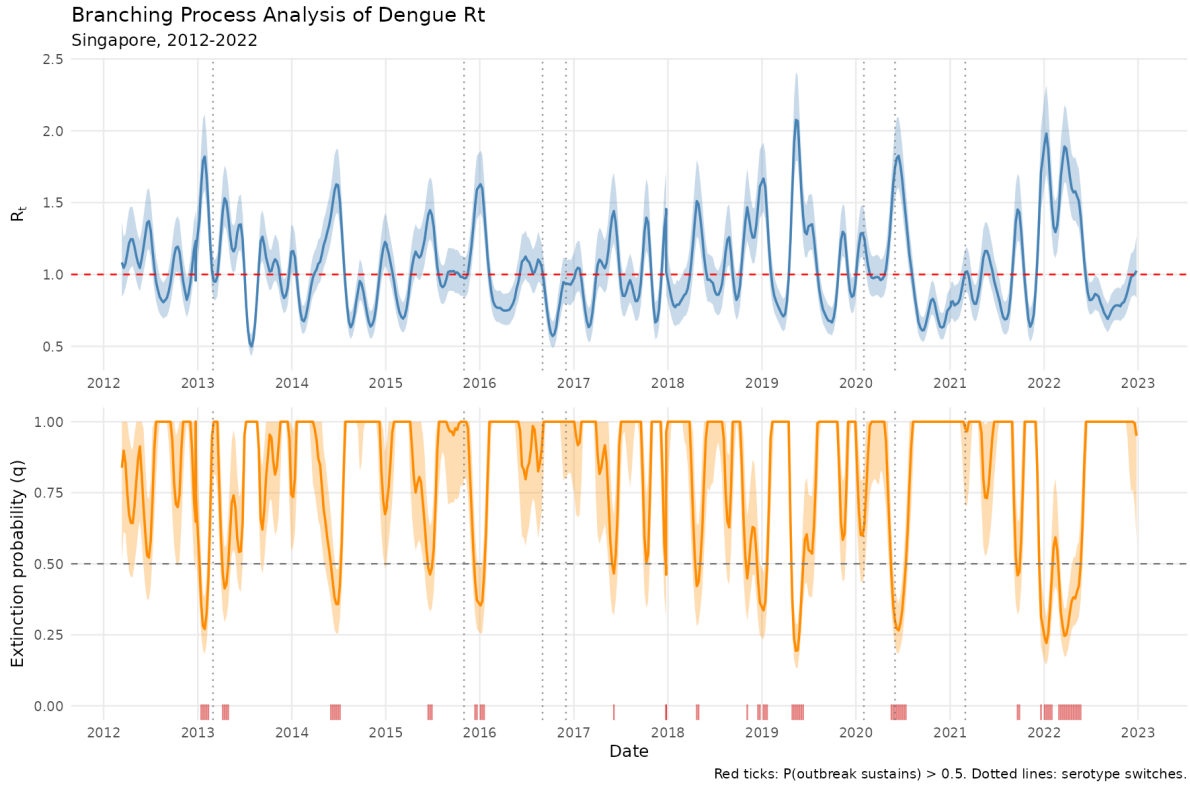

Figure 10: Branching process analysis. **Top:** Posterior median R_t trajectory with 95% credible band. **Bottom:** Time-varying outbreak sustaining probability $P(\text{sustain}) = 1 - q$, where q is the extinction probability under the negative binomial offspring distribution. Vertical dashed lines mark serotype switch events. The strong anticorrelation ($\rho_{\text{Spearman}} = -0.94$) between R_t and extinction probability confirms that small changes in R_t translate to large changes in outbreak risk.

3.6 VAE Comparison: A Meaningful Negative Result

The VAE trained on raw case counts does not discover the GP residual structure (Figure 12). The maximum Spearman correlation between any latent dimension and f_{residual} is 0.09 ($p = 0.035$); PCA components of the latent space show correlations near zero with all epidemiological features. Reconstruction error does not spike at serotype switch dates (Mann–Whitney $p = 0.97$).

This negative result validates the renewal equation framework. The mechanistic model structure (renewal equation + GP) captures biologically meaningful temporal variation that a purely data-driven approach cannot discover from case counts alone. The GP residual encodes epidemiological information about transmission dynamics—not just statistical smoothing of the case time series.

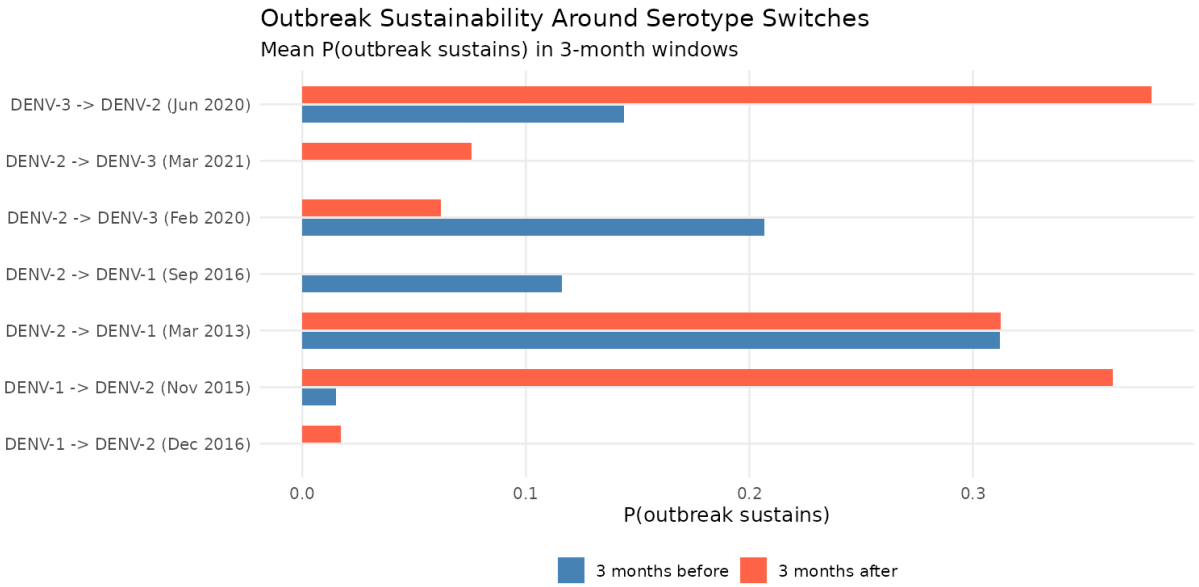

Figure 11: Outbreak sustaining probability before (grey) versus after (coloured) each serotype switch, computed as the mean $P(\text{sustain})$ in 3-month windows. Five of 7 switches show increased vulnerability post-switch (mean increase: +6 percentage points), consistent with the immunity-depletion mechanism.

3.7 Spectral Analysis: Multi-Scale Temporal Structure

The residual GP contains rich multi-scale temporal structure even after climate removal (Figure 13). The dominant periodicities and their epidemiological interpretations are:

- **26 weeks (6 months):** Strongest peak, reflecting the biannual dengue epidemic pattern driven by Singapore's two monsoon seasons (Northeast: Dec–Mar, Southwest: Jun–Sep).
- **82 weeks (1.6 years):** Multi-year cycling beyond annual seasonality.
- **115 weeks (2.2 years):** Consistent with theoretical predictions of serotype replacement cycling [Wearing and Rohani, 2006].
- **191 weeks (3.7 years):** Near the theoretical 4-serotype rotation period.
- **52 weeks (1 year):** Residual annual seasonality not captured by temperature and rainfall, possibly reflecting non-climatic seasonal drivers (school terms, travel patterns, vector control campaign timing).

The presence of a 52-week peak in the *residual* (after climate covariates are removed) indicates that the temperature and rainfall covariates do not fully capture all seasonal drivers of dengue transmission.

3.8 Counterfactual Analysis: Quantifying Excess Cases

The February 2020 DENV-3 emergence is the only serotype switch producing clearly significant excess cases: +4,419 (95% CrI: 2,925 to 6,043) over the 6 months post-switch (Figure 14). Over the full decade, positive and negative excursions approximately cancel (net median: −763 cases; 95% CrI: −12,129 to +10,556).

Most switches show excess cases indistinguishable from zero, suggesting that only *novel serotype introductions*—not rotations between familiar serotypes—produce detectable excess transmission at

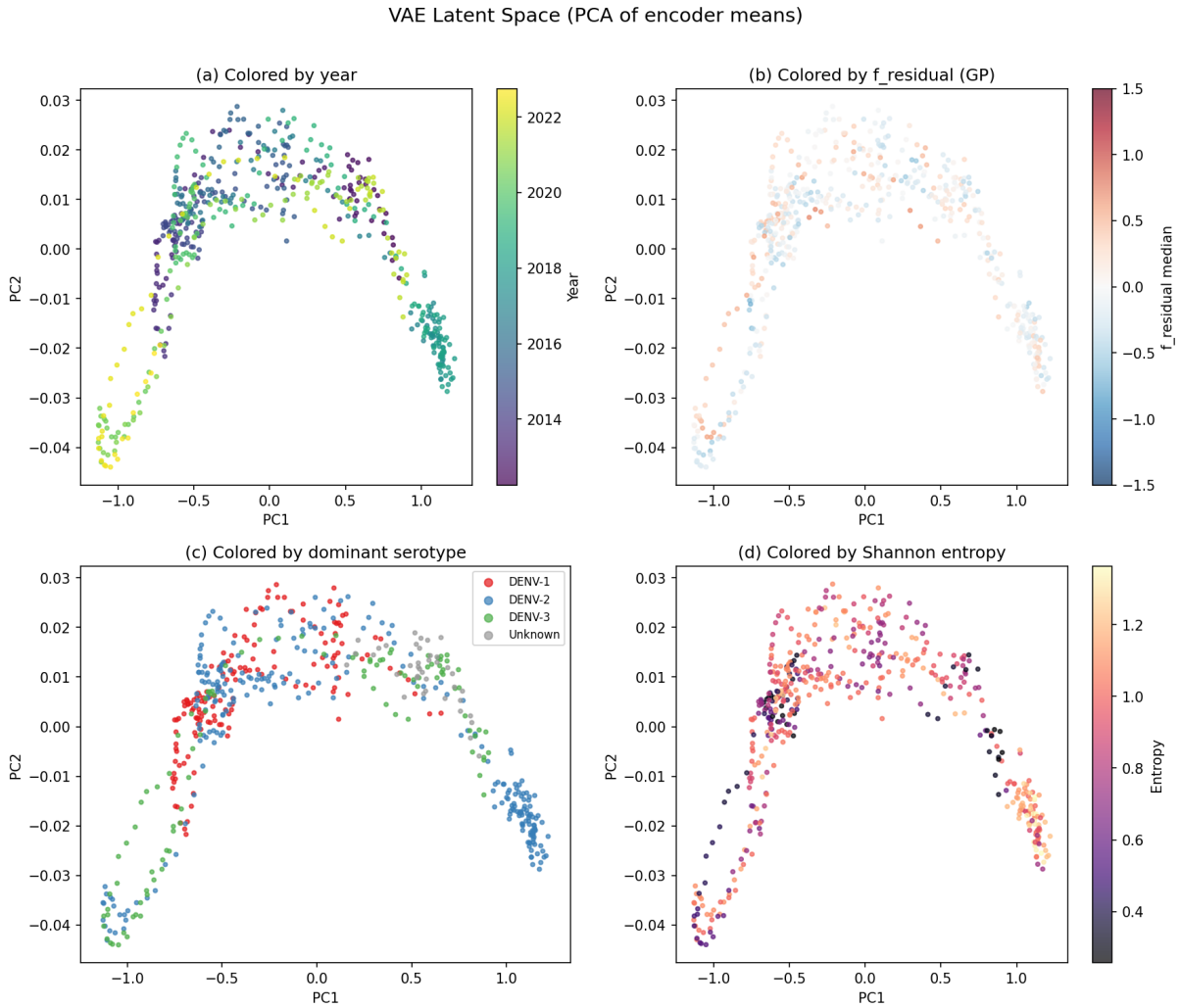

Figure 12: VAE latent space visualised via PCA, coloured by (a) time period, (b) posterior mean f_{residual}, (c) dominant serotype, and (d) Shannon entropy. No clustering by any epidemiological feature is apparent. Maximum Spearman $\rho = 0.09$ between any latent dimension and f_{residual}, confirming that the VAE does not recover the structure that the renewal equation + GP framework captures.

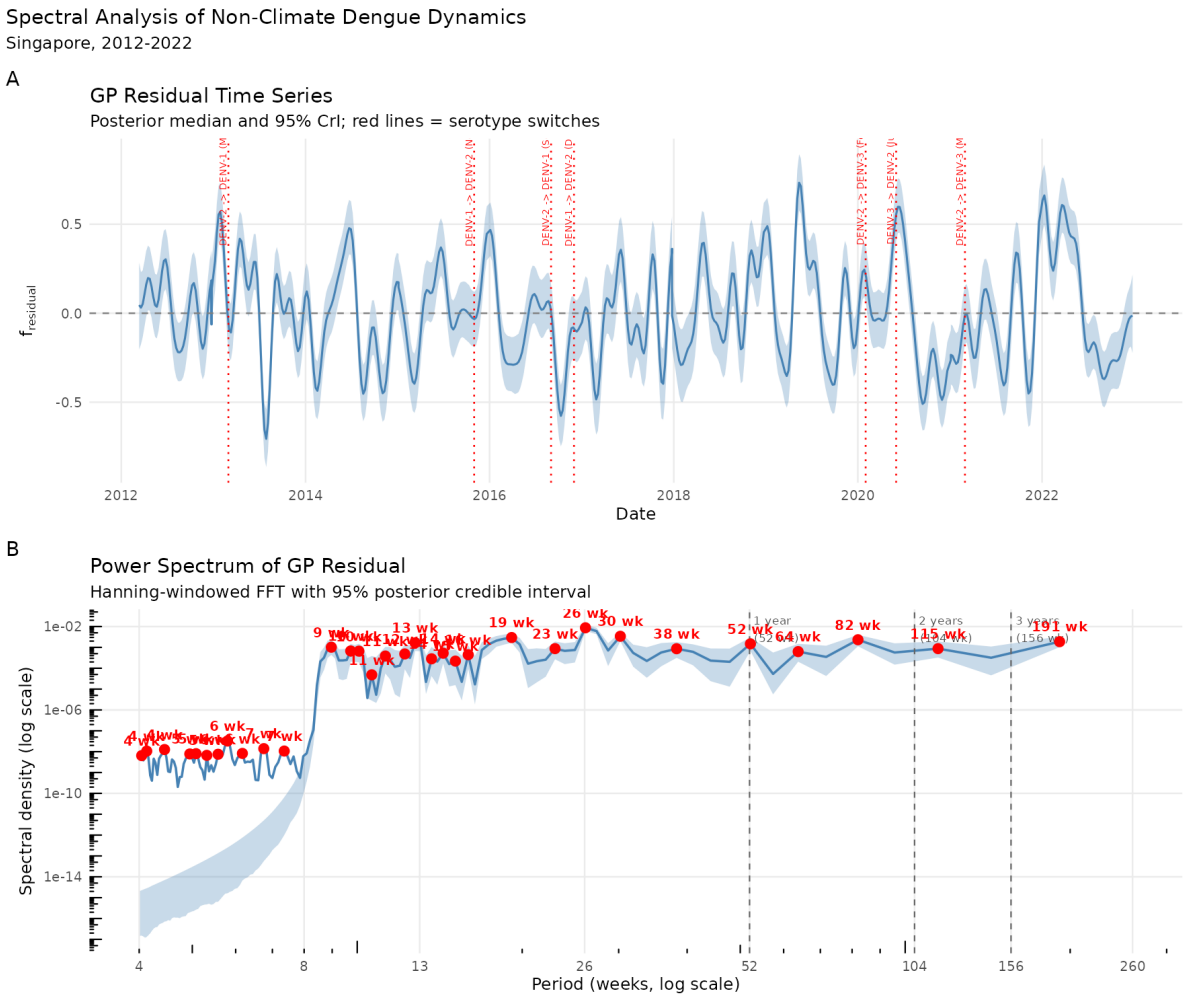

Figure 13: Power spectrum of the posterior mean f_{residual} computed via Lomb–Scargle periodogram. Vertical annotations mark reference periods corresponding to known dengue dynamics. The spectrum reveals multi-scale temporal structure from sub-annual (13–26 weeks) through multi-year (82–191 weeks) timescales, consistent with theoretical predictions of oscillations driven by cross-immunity between 4 co-circulating serotypes [Wearing and Rohani, 2006]. Shaded band shows 95% credible interval from posterior draws.

the national level. The September 2016 DENV-2→DENV-1 and March 2021 DENV-2→DENV-3 switches show marginally significant *reductions* in cases (−693 and −296 respectively), potentially reflecting post-epidemic troughs.

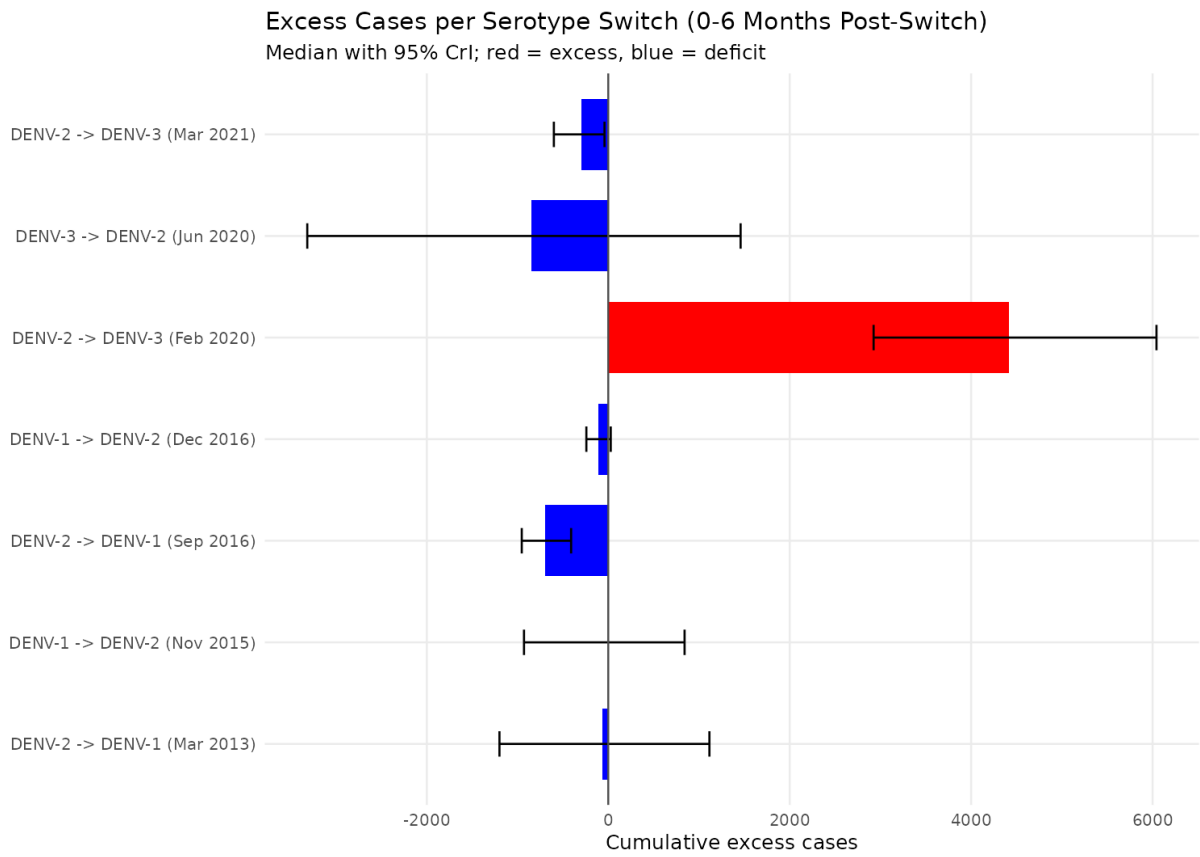

Figure 14: Excess cases attributable to non-climate drivers per serotype switch event, computed as cumulative observed minus counterfactual cases over 6 months post-switch. Points show posterior median; error bars show 95% CrI. Only the February 2020 DENV-3 emergence (red) produces a significant positive excess, though this event is confounded with COVID-19 behavioural changes that may have increased domestic *Aedes* exposure.

4 Discussion

4.1 Summary of Findings

This study demonstrates that a semiparametric Bayesian framework—combining a renewal equation with an additive GP residual—can effectively decompose dengue R_t into measured and unmeasured components. Climate is genuinely negligible at weekly resolution (0.4% of variance). The residual GP captures multi-scale temporal structure consistent with serotype-driven immunity dynamics, including periodicities at 2.2 and 3.7 years matching theoretical predictions [Wearing and Rohani, 2006]. Five of 7 serotype switches associate with elevated R_t, and the branching process analysis shows these create measurable windows of outbreak vulnerability.

However, the framework has clear limitations. The GP absorbs any smooth temporal variation, making causal attribution to serotype dynamics inherently ambiguous. The GP provides no forecasting skill beyond naive persistence. And the VAE comparison, while validating the

mechanistic framework, also highlights the small effect sizes involved (maximum $\rho = 0.09$).

4.2 Comparison with Existing Literature

Our finding that climate is a weak predictor of weekly dengue R_t is consistent with Finch et al. [2025], who found that climate explains only a modest fraction of dengue dynamics in Singapore after accounting for serotype competition. The serotype-immunity mechanism we describe is well-established theoretically [Reich et al., 2013] but has rarely been quantified through GP decomposition of the R_t time series.

The spectral analysis revealing ~2-year and ~4-year cycles is consistent with the theoretical prediction of multi-year oscillations driven by cross-immunity between 4 co-circulating serotypes [Wearing and Rohani, 2006]. The 6-month dominant periodicity likely reflects the bimodal seasonal pattern driven by Singapore's two monsoon seasons.

4.3 Limitations

1. **Climate decomposition is scale-dependent.** The finding that climate explains $<$1% of R_t variance applies specifically at weekly resolution. Climate likely operates through seasonal modulation of vector populations on longer timescales. A model with explicit seasonal kernels or monthly aggregation might attribute more variance to climate.

2. **GP dominance is partly structural.** With an estimated length scale of ~3 weeks, the GP can track any smooth temporal variation, whether or not it originates from serotype dynamics. The 99.5% variance share is a consequence of the model giving the GP sufficient flexibility to absorb nearly all signal—it does not constitute evidence that the residual is *caused by* serotype dynamics.

3. **Serotype associations are descriptive with low statistical power.** With only 7 switch events in 10 years, formal hypothesis testing is not meaningful. The confounding of the 2020 DENV-3 switch with COVID-19 further complicates attribution.

4. **Case ascertainment is assumed constant.** While a constant reporting fraction cancels in the renewal equation [Cori et al., 2013], serotype-dependent ascertainment could bias the post-hoc serotype analysis. Tan et al. [2019] estimate infection-to-notification ratios ranging from 3:1 to 16:1 depending on serotype and period. Time-varying ascertainment is not identifiable from case data alone.

5. **Forecasting failure is fundamental.** The GP's inability to beat a random walk baseline reflects the inherent limitation of a smoothing prior that contains no mechanistic forward model, not a tuning problem.

6. **Spatial aggregation.** The model treats Singapore as a single spatial unit, potentially masking heterogeneity in neighbourhood-level transmission driven by local vector densities and demographic structure.

7. **Generation interval uncertainty.** The discretised generation interval is fixed, not estimated. While sensitivity analysis across temperature-dependent EIP scenarios (17–25 day total GI) shows qualitatively stable results, jointly estimating the generation interval could improve inference.

8. **Single time series.** The 11-year observation window provides limited statistical power for characterising multi-year cycles (e.g., the 3.7-year peak in the spectral analysis corresponds to fewer than 3 complete cycles).

4.4 Future Directions

Three extensions would address the most important limitations. First, integrating the GP decomposition with an explicit multi-serotype susceptible-infected-recovered (SIR) model would enable causal attribution to serotype dynamics rather than post-hoc pattern matching. Second, spatial disaggregation using dengue cluster data could reveal neighbourhood-level drivers masked by national aggregation. Third, hybrid approaches combining GP retrospective decomposition with mechanistic forward models could potentially achieve both accurate decomposition and operationally useful forecasting.

4.5 Conclusions

The Bayesian renewal equation with GP residual provides a principled framework for decomposing dengue R_t into climate-attributable and residual components. Climate is genuinely negligible at weekly resolution. The residual GP captures multi-scale temporal structure consistent with serotype-driven immunity dynamics, but the framework has fundamental limitations for causal decomposition (the GP absorbs any smooth variation) and operational forecasting (the GP provides no predictive skill beyond persistence). The cross-kernel comparison confirms robustness. The VAE comparison validates that the mechanistic model structure extracts epidemiologically meaningful information that purely data-driven methods miss. Future work should integrate GP decomposition with explicit serotype-immunity models to enable both causal attribution and operational forecasting.

Declaration on Use of AI Tools

Claude (Anthropic) was used as a coding assistant for Stan model development, R/Python analysis scripts, and to assist with drafting and editing this report. All scientific decisions, model specification choices, and interpretation of results were made by the author. Code and analysis were reviewed and validated by the author.

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